

PoC → TRANSLATIONAL RESEARCH

Real World Research → Proof of Concept +
Translational Research

Effective contact tracing effort

key elements → key steps

- 1) case investigation + contact information
- 2) contact listing
- 3) "follow up"

CONTACT TRACING EBOLA

key challenges identified

- incomplete identification of contacts
- inefficiencies in the paper based reporting system
- inadequate data collection
- transcription errors
- delays in each contact tracing step

Research — evaluate effectiveness of a smartphone-based electronic data capture and management system relative to the standard paper based system
for Ebola ← Contact tracing
monitoring

Assess operational issues

Clustered randomised trial

Design → changed to a proof-of-concept demonstration project in early June 2015 (due to decline in Ebola cases)

ECTapp

- Stage 1) Registration Stage
- 2) Assignment stage
- 3) Visitation stage

Alert system — through SMS system

BA Ebola

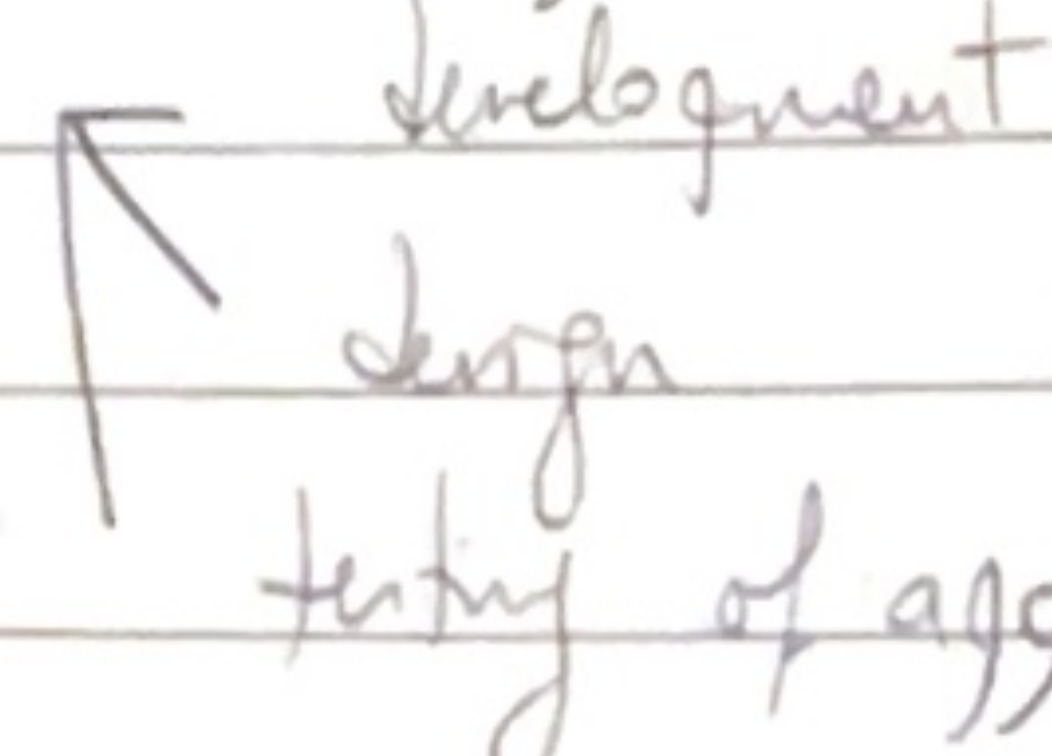
Surveillance

- contact tracing importance
mHealth approaches in other cases - apps

Methods

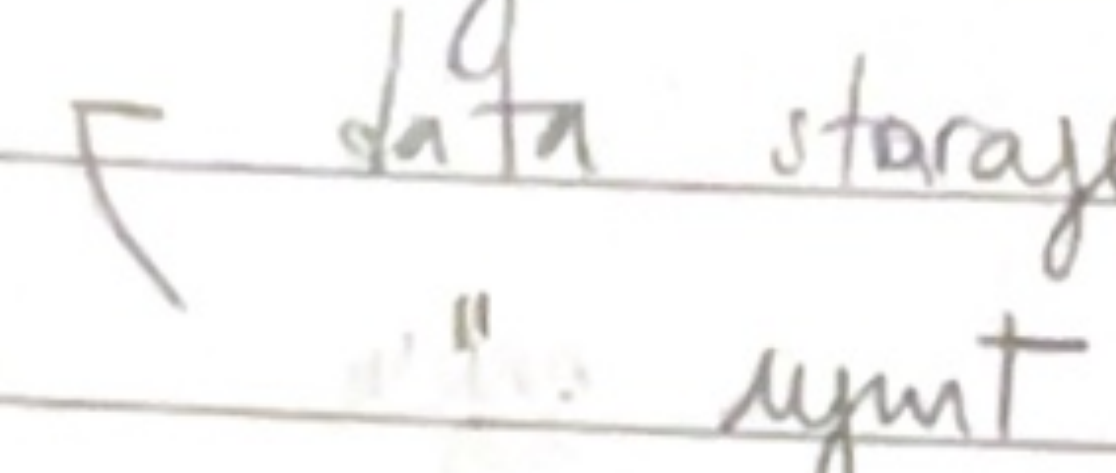
Original Study → P.C

qualitative methods for process evaluation

ECT app — d/ 

planned introduction of ECT app

study training

issues 

Data analysis

key results

discuss paper based vs ECT app

process evaluation → qualitative results
suggested improvements

Strengths and limitations of both approaches

Key Study challenges

- Ascertaining the contact tracing process
- Obtaining up to date info on confounding factors
- Achieving intended sample size
- Changing from a cluster randomised trial to a RCT study

↓
gold standard to biases → lost

ECT app designed to error free based

Main development challenges

Working within a rapidly shifting situation
reflecting complex design structure is the technical design of the app

Speed with which app needed to be developed

App structure - tech changes in short time

limited time for pre and pilot testing

adapting users with little or no technological expertise

Implementation issues

Mitigating risks to the contact tracing and surveillance process
phased introduction of the intervention

- poor network coverage
- technical
- geographical + infrastructural challenges
- short battery life and charging issues
- time lag between training and emergency
- assignment of large units of ECR contacts using app

Mitigation action

testing phones

providing spare batteries

having field monitor

strategies to address issues with the app design

→ Process evaluation

process indicators ← activities
outputs

in-house → semi structured interviews — personnel

independent evaluation ← ext partners
semi struct interviews
observations of app being used

Results — staff preferred the app → reasons (...)

Changes she'd do

→ simplify app

→ improve battery life

→ tech features

⇒ Ben Glanvason for
gull info out → researchers

work with
PM, Judges, architects, info governance (regulations)
living w/ patients
curating data
collaborative w/ UCLH
other
centres in UK
Lungs
Oxford
Cambridge

WRITING

+

GOVERNANCE

DATA

check

⇒ Digital alert for Sepsis (marginal)

can check material + video again